4 **(a)** Sound waves are longitudinal waves. State what is meant by a *longitudinal* wave.

.....

..... [1]

(b) A stationary sound wave in air has amplitude A. In an experiment, a detector is used to determine A^2. The variation of A^2 with distance x along the wave is shown in Figure 4.1.

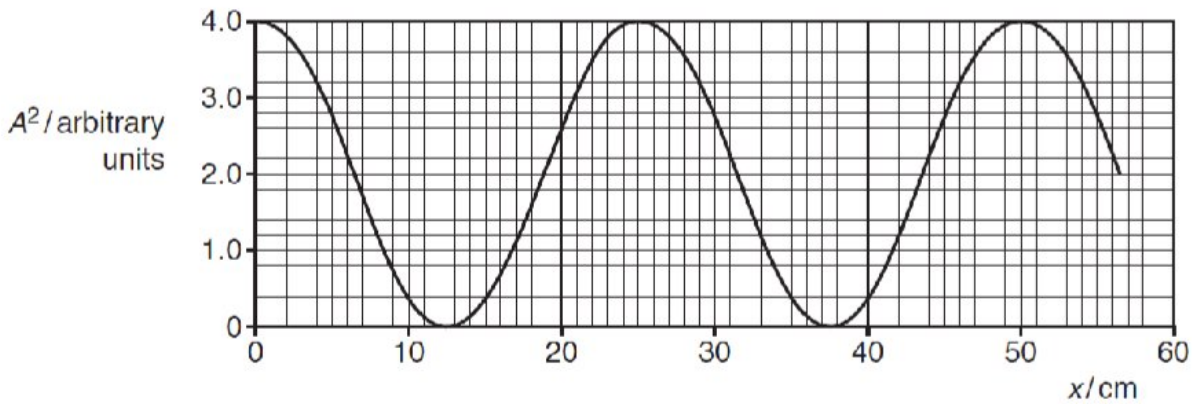


Figure 4.1

(i) State the phase difference between the vibrations of an air particle at x = 25 cm and the vibrations of an air particle at x = 50 cm.

phase difference = ... [1]

(ii) The speed of the sound in the air is 330 m s^{-1}. Determine the frequency of the sound wave.

frequency = ... Hz [2]

(iii) Determine the ratio

$$\frac{\text{intensity of wave at } x = 20\,\text{cm}}{\text{intensity of wave at } x = 25\,\text{cm}}$$

ratio = ... [2]

[Total: 6]